Veer Singh

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Education

University of Illinois Urbana-Champaign **Aug 2024 – May 2028**
B.S. Computer Science & Statistics | GPA: 3.8; Major GPA: 3.9 *Champaign, IL*

- **Relevant Coursework:** Operating Systems, Computer Architecture, Systems Programming, Distributed Systems, Algorithm Design, Parallel GPU Programming, Cloud Networking, Machine Learning

Experience

Balyasny Asset Management **Jun 2027 – Aug 2027**
Incoming Quantitative Developer Intern, Systematic *New York, NY*

DRW **Jan 2027 – May 2027**
Software Engineer Intern, Market Data *Chicago, IL*

National Center for Supercomputing Applications (NCSA) **Jun 2026 – Present**
SPIN Supercomputing Research Assistant *Champaign, IL*

- Building LLM-as-a-Service (LLMaaS) infrastructure on the NCSA DeltaAI supercomputer (608x H100 GPUs), enabling multi-tenant transformer model serving across vLLM and TensorRT-LLM backends at rack scale.
- Profiled concurrent 70B+ parameter inference workloads using Nsight Systems, Slurm, and containerized deployments; reduced end-to-end P95 latency by ∼30% by addressing memory-bandwidth bottlenecks and migrating to FP8 quantization.

Perplexity **Mar 2026 – May 2026**
AI Advisors Fellow *Champaign, IL*

- Scoped and deployed AI workflow automations for local businesses via Perplexity Computer under the Comet Activator Fellows Program, translating business requirements into executable workflows and integrating external APIs.
- Evaluated workflow reliability using task-completion and human-escalation rates to identify automation gaps and inform deployment improvements.

Handshake **Sep 2025 – Dec 2025**
Model Validation, RLHF Co-Op *Remote*

- Created and evaluated training data for OpenAI's RLHF pipelines via the Feather Platform, assessing model outputs for correctness, instruction-following, safety, and response quality; produced preference annotations used to fine-tune model behavior and support reward-model training.

University of Illinois Urbana-Champaign **Aug 2025 – Jan 2026**
Course Assistant – CS340: Computer Systems *Champaign, IL*

- Led office hours for 100+ students on OS fundamentals: memory hierarchy, virtual memory, process scheduling, file I/O, and TCP/IP socket programming in C.
- Diagnosed segmentation faults, race conditions, and deadlocks in multithreaded C programs using gdb, valgrind, strace, and perf; authored Linux kernel process-scheduling analyses from live tracing experiments.

Projects & Contributions

Clad – Google Summer of Code 2026 | *C++23, LLVM/Clang, CUDA, OpenMP, CMake* **Apr 2026 – Present**

- Contributing to Clad, an LLVM/Clang plugin for source-to-source automatic differentiation (AD) of C++ programs.
- Implemented AD support for parallelized C++ programs by extending Clad's AST traversal and derivative code generation passes; added regression and edge-case test coverage for reverse-mode AD across loop and conditional constructs.

Voxel-Based 3D Tumor Segmentation | *CUDA, PyTorch, Python* **Sep 2023 – May 2024**

- Designed a context-aware voxel sampling pipeline for volumetric tumor segmentation using a hybrid Point-Voxel CNN built on top of nnU-Net.
- Implemented custom CUDA data-loading and preprocessing kernels for volumetric sampling; awarded Grand Prize at Regeneron ISEF 2024 ($7,500 scholarship) – top honor among 1,700+ projects from 49 countries.

Technical Skills

Languages: C++ (17/20/23), C, Python, Rust, SQL, Kotlin
HPC / Parallel: CUDA, OpenMP, MPI, pthreads, ROCm, Slurm, TensorRT, TensorRT-LLM, vLLM, Nsight Systems/Compute
DevOps / Systems: CMake, GCC, gdb, valgrind, strace, perf, Docker, Kubernetes, AWS, GitHub Actions, CI/CD, Linux
Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, FastAPI

Honors & Leadership

Regeneron ISEF Grand Awardee : top honor among 1,700+ projects, $7,500 scholarship | Akuna Trading Challenge 2026 | HackIllinois Finalist | Citadel Travel Fellow | Optiver Futurefocus | GSOC | PennApps Finalist | Illinois Microtech Exec Board | HackIllinois Exec Board | AIMO Qualifier | RMO Finalist